

Assignment: Building an Advanced Ensemble Model with Feature Scaling & Deployment

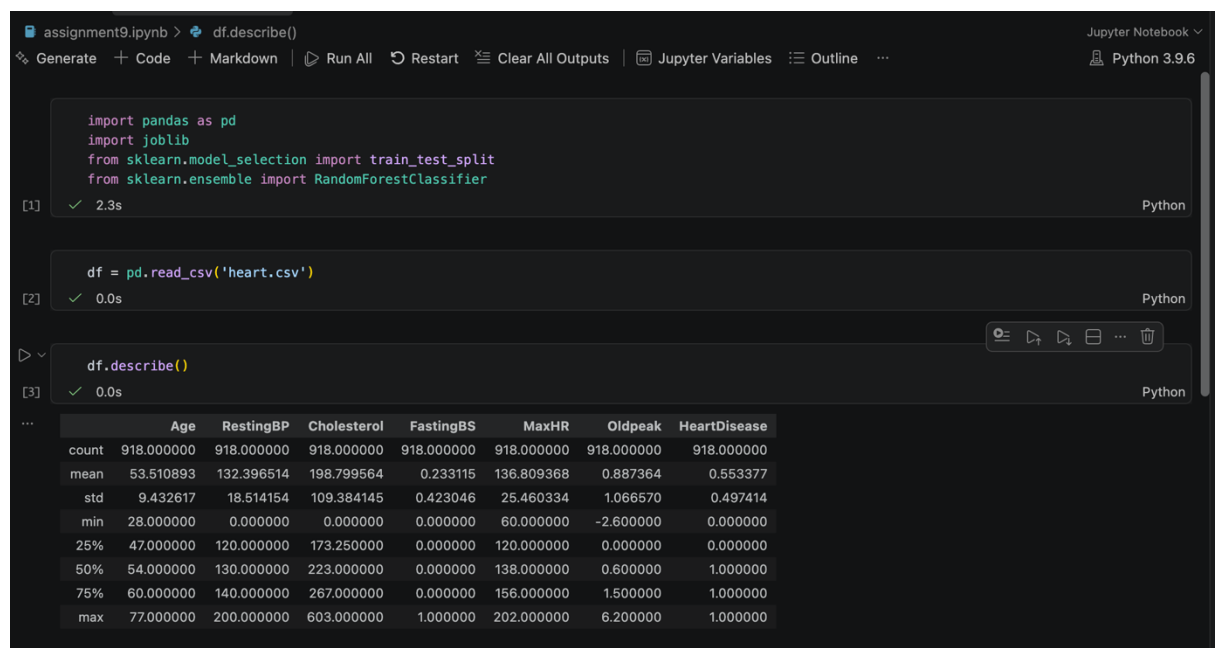
Sudarshan Devadiga

150096724089

Objective: Train a Gradient Boosting ensemble classifier on the Kaggle Heart Failure Prediction dataset, handle categorical feature encoding and continuous variable scaling, serialize model artifacts, and build a risk assessment web interface using Streamlit.

1. Library Imports & Dataset Inspection

- **Libraries:** Loaded pandas, sklearn (GradientBoostingClassifier, train_test_split, StandardScaler), and joblib for data preparation, modeling, and serialization.
- **Dataset Overview:** Imported heart.csv containing 918 records across 12 feature columns: Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope, and the target HeartDisease.
- **Data Quality:** Verified dataset shape (918, 12) with zero missing values.



```
assignment9.ipynb > df.describe()
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline | Python 3.9.6
```

```
[1] ✓ 2.3s Python
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
```

```
[2] ✓ 0.0s Python
df = pd.read_csv('heart.csv')
```

```
[3] ✓ 0.0s Python
df.describe()
```

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisease
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	0.233115	136.809368	0.887364	0.553377
std	9.432617	18.514154	109.384145	0.423046	25.460334	1.066570	0.497414
min	28.000000	0.000000	0.000000	0.000000	60.000000	-2.600000	0.000000
25%	47.000000	120.000000	173.250000	0.000000	120.000000	0.000000	0.000000
50%	54.000000	130.000000	223.000000	0.000000	138.000000	0.600000	1.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.500000	1.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000	6.200000	1.000000

2. Feature Selection & Preprocessing

- **Target Separation:** Separated target label HeartDisease from the feature matrix.
- **Categorical Encoding:** Applied pd.get_dummies(drop_first=True) to transform categorical string columns into numeric binary variables.
- **Numerical Scaling:** Introduced a StandardScaler pipeline to normalize continuous features (Age, RestingBP, Cholesterol, MaxHR, Oldpeak) to

optimize the boosting algorithm. Saved the feature columns as `feature_names.pkl` and the fitted scaler as `scaler.pkl`.

```
[4] X = df.drop(columns=['HeartDisease'])
    y = df['HeartDisease']
    ✓ 0.0s Python

[5] X = pd.get_dummies(X, drop_first=True)
    ✓ 0.0s Python

[6] model_columns = list(X.columns)
    joblib.dump(model_columns, 'model_columns.pkl')
    ✓ 0.0s Python
... ['model_columns.pkl']
```

3. Ensemble Model Training & Serialization

- **Dataset Splitting:** Partitioned the dataset into 75% training and 25% testing sets using `random_state=73`.
- **Boosting Ensemble:** Trained a `GradientBoostingClassifier` with 100 decision trees (`n_estimators=100`) and a learning rate of 0.1.
- **Model Export:** Serialized the trained ensemble estimator into `gb_classifier.pkl` using `joblib`.

```
[7] X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    ✓ 0.0s Python

[8] model = RandomForestClassifier(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)
    ✓ 0.1s Python
...
    RandomForestClassifier
    RandomForestClassifier(random_state=42)

[9] joblib.dump(model, 'ensemble_model.pkl')
    print("Model trained and saved successfully!")
    ✓ 0.0s Python
... Model trained and saved successfully!
```

4. Deployment Interface (app.py) Built a Streamlit web application titled "Cardiovascular Risk Assessment Tool". The script dynamically captures user inputs, applies dummy encoding, and routes continuous variables through `scaler.pkl` before aligning features against `feature_names.pkl`. Output states are routed to display "Alert: Elevated Cardiovascular Risk Detected" or "Result: Low Cardiovascular Risk."

```
1 import streamlit as st
2 import pandas as pd
3 import joblib
4
5 # Load trained model, expected feature columns, and the scaler
6 model = joblib.load('gb_classifier.pkl')
7 model_columns = joblib.load('feature_names.pkl')
8 scaler = joblib.load('scaler.pkl')
9
10 st.title("Cardiovascular Risk Assessment Tool")
11
12 # Collect ALL inputs present in the raw dataset
13 age = st.number_input("Age", min_value=1, max_value=120, value=50)
14 sex = st.selectbox("Sex", ["M", "F"])
15 chest_pain = st.selectbox("Chest Pain Type", ["TA", "ATA", "NAP", "ASY"])
16 resting_bp = st.number_input("Resting Blood Pressure", min_value=50, max_value=200, value=120)
17 cholesterol = st.number_input("Cholesterol", min_value=100, max_value=600, value=200)
18 fasting_bs = st.selectbox("Fasting Blood Sugar > 120 mg/dl", [0, 1])
19 resting_ecg = st.selectbox("Resting ECG", ["Normal", "ST", "LWV"])
20 max_hr = st.number_input("Max Heart Rate", min_value=60, max_value=220, value=150)
21 exercise_angina = st.selectbox("Exercise Induced Angina", ["N", "Y"])
22 oldpeak = st.number_input("Oldpeak", value=0.0)
23 st_slope = st.selectbox("ST Slope", ["Up", "Flat", "Down"])
24
25 if st.button("Assess Risk"):
26     # Convert user input into a single-row DataFrame
27     input_dict = {
28         'Age': age, 'Sex': sex, 'ChestPainType': chest_pain,
29         'RestingBP': resting_bp, 'Cholesterol': cholesterol,
30         'FastingBS': fasting_bs, 'RestingECG': resting_ecg,
31         'MaxHR': max_hr, 'ExerciseAngina': exercise_angina,
32         'Oldpeak': oldpeak, 'ST_Slope': st_slope
33     }
34     input_df = pd.DataFrame([input_dict])
35
36     # Apply dummy encoding
37     input_df_encoded = pd.get_dummies(input_df)
38
39     # Reindex to match model features and fill missing dummies with 0[cite: 1]
40     input_df_encoded = input_df_encoded.reindex(columns=model_columns, fill_value=0)
41
42     # Scale the continuous numerical columns
43     num_cols = ['Age', 'RestingBP', 'Cholesterol', 'MaxHR', 'Oldpeak']
44     input_df_encoded[num_cols] = scaler.transform(input_df_encoded[num_cols])
45
46     # Predict[cite: 1]
47     prediction = model.predict(input_df_encoded)
48
49     if prediction[0] == 1:
50         st.error("Alert: Elevated Cardiovascular Risk Detected.")
51     else:
52         st.success("Result: Low Cardiovascular Risk.")
```

5. Deployment Execution & UI Metrics

Metric / Deployment Spec	Value / Output
Framework	Streamlit
Command	streamlit run app.py
Local Endpoint	http://localhost:8501
Base Classifier	Gradient Boosting Ensemble (100 Estimators)
Artifacts Loaded	3 (gb_classifier.pkl, feature_names.pkl, scaler.pkl)

```

Problems Output Debug Console Terminal Ports Jupyter
sudarshandevadiga@sudarshans-MacBook-Air Assignment 9 % python3 -m streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.20.10.3:8501

For better performance, install the Watchdog module:

$ xcode-select --install
$ pip install watchdog

^C Stopping...
sudarshandevadiga@sudarshans-MacBook-Air Assignment 9 % clear
sudarshandevadiga@sudarshans-MacBook-Air Assignment 9 % python3 -m streamlit run app.py

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```

A screenshot of a web browser window displaying a 'Cardiovascular Risk Assessment Tool'. The browser's address bar shows 'localhost:8501'. The tool's interface is dark-themed and features several input fields with labels and values: 'Age' (50), 'Sex' (M), 'Chest Pain Type' (TA), 'Resting Blood Pressure' (120), 'Cholesterol' (200), 'Fasting Blood Sugar > 120 mg/dl' (0), and 'Resting ECG' (Normal). Each input field has a dropdown arrow on its right side. In the top right corner of the browser window, there is a 'Deploy' button and a 'School' icon. The title of the tool, 'Cardiovascular Risk Assessment Tool', is prominently displayed in the center of the page.

Streamlit

Ask Gemini

localhost:8501

Deploy

Fasting Blood Sugar > 120 mg/dl

0

Resting ECG

Normal

Max Heart Rate

150

-

+

Exercise Induced Angina

N

Oldpeak

0.00

-

+

ST Slope

Up

Assess Risk

Result: Low Cardiovascular Risk.